

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC 95% CI	Samples	Positive Rate	Cross-Val Folds
Attention Pool Extra3	A Lines	Random Forest	0.758	[0.726-0.790]	43689	0.472	5
Attention Pool Extra3	A Lines	Logistic Regression	0.795	[0.778-0.813]	43689	0.472	5
Attention Pool Extra3	A Lines	XGBoost	0.794	[0.779-0.810]	43689	0.472	5
Attention Pool Extra3	Large Consolidation	Random Forest	0.863	[0.847-0.880]	43689	0.110	5
Attention Pool Extra3	Large Consolidation	Logistic Regression	0.919	[0.913-0.924]	43689	0.110	5
Attention Pool Extra3	Large Consolidation	XGBoost	0.901	[0.886-0.915]	43689	0.110	5
Attention Pool Extra3	Pleural Effusion	Random Forest	0.674	[0.602-0.745]	43689	0.011	5
Attention Pool Extra3	Pleural Effusion	Logistic Regression	0.808	[0.759-0.856]	43689	0.011	5
Attention Pool Extra3	Pleural Effusion	XGBoost	0.829	[0.778-0.880]	43689	0.011	5
Attention Pool Extra3	Other Pathology	Random Forest	0.644	[0.614-0.674]	43689	0.405	5
Attention Pool Extra3	Other Pathology	Logistic Regression	0.690	[0.662-0.717]	43689	0.405	5
Attention Pool Extra3	Other Pathology	XGBoost	0.684	[0.662-0.707]	43689	0.405	5
Attention Pool Extra2	A Lines	Random Forest	0.719	[0.651-0.787]	43689	0.472	5
Attention Pool Extra2	A Lines	Logistic Regression	0.762	[0.721-0.802]	43689	0.472	5
Attention Pool Extra2	A Lines	XGBoost	0.761	[0.717-0.806]	43689	0.472	5
Attention Pool Extra2	Large Consolidation	Random Forest	0.819	[0.760-0.879]	43689	0.110	5
Attention Pool Extra2	Large Consolidation	Logistic Regression	0.899	[0.880-0.918]	43689	0.110	5
Attention Pool Extra2	Large Consolidation	XGBoost	0.866	[0.817-0.916]	43689	0.110	5
Attention Pool Extra2	Pleural Effusion	Random Forest	0.592	[0.553-0.631]	43689	0.011	5
Attention Pool Extra2	Pleural Effusion	Logistic Regression	0.771	[0.723-0.820]	43689	0.011	5
Attention Pool Extra2	Pleural Effusion	XGBoost	0.756	[0.643-0.868]	43689	0.011	5
Attention Pool Extra2	Other Pathology	Random Forest	0.601	[0.577-0.626]	43689	0.405	5
Attention Pool Extra2	Other Pathology	Logistic Regression	0.662	[0.635-0.689]	43689	0.405	5
Attention Pool Extra2	Other Pathology	XGBoost	0.650	[0.617-0.682]	43689	0.405	5
Attention Pool Extra1	A Lines	Random Forest	0.710	[0.692-0.728]	43689	0.472	5
Attention Pool Extra1	A Lines	Logistic Regression	0.751	[0.737-0.766]	43689	0.472	5
Attention Pool Extra1	A Lines	XGBoost	0.755	[0.742-0.767]	43689	0.472	5
Attention Pool Extra1	Large Consolidation	Random Forest	0.824	[0.801-0.847]	43689	0.110	5
Attention Pool Extra1	Large Consolidation	Logistic Regression	0.884	[0.871-0.897]	43689	0.110	5
Attention Pool Extra1	Large Consolidation	XGBoost	0.871	[0.850-0.892]	43689	0.110	5
Attention Pool Extra1	Pleural Effusion	Random Forest	0.655	[0.624-0.685]	43689	0.011	5
Attention Pool Extra1	Pleural Effusion	Logistic Regression	0.785	[0.738-0.832]	43689	0.011	5
Attention Pool Extra1	Pleural Effusion	XGBoost	0.815	[0.773-0.856]	43689	0.011	5
Attention Pool Extra1	Other Pathology	Random Forest	0.605	[0.592-0.617]	43689	0.405	5
Attention Pool Extra1	Other Pathology	Logistic Regression	0.655	[0.639-0.671]	43689	0.405	5
Attention Pool Extra1	Other Pathology	XGBoost	0.645	[0.640-0.650]	43689	0.405	5